

WEEK 3-PROJECT SUMMARY

What you planned to accomplish.

The objective of this week was to transform Week 2 analysis into a more complete and decision-oriented business intelligence solution. Week 3 now focused on: Develop → Improve → Validate → Prepare for Final Integration.

The deliverables were:

- Extend the SQL and Python analysis.
- Expand the dashboard.
- Validate findings across tools.
- Write recommendations.

What you completed.

- 6 SQL analyses
- 5 Python analyses
- a [4]-page Power BI dashboard
- KPI analysis
- 5 validated findings
- 5 recommendations
- the validation log

Major development activities.

The analysis covered:

- Customer segment performance analysis.
- Transaction trends over months.
- International versus Domestic activity.
- Risk review analysis

Key findings/results.

- Transactions fell 9.65% in February and recovered in March.
- Risk-review rate was 40.54% at NGN 200K and above, compared with 17.66% below 10K.
- POS Bill Payment (7.22%) and USSD Transfer (7.20%) had the highest failure rates.
- SME had the highest value per customer (NGN 387,923) and Student the highest frequency (8.23).

Testing and validation performed.

- SQL and Python outputs reconciled on counts and rates.
- Dashboard KPI cards were checked against totals (12,000 transactions, NGN 560.48M).
- Refer to your validation log for the evidence.

Improvements made.

- In SQL; Extended the SQL work with CTEs, window functions, ranking, month-over-month comparisons, customer-level analysis and deeper combinations such as Channel x Transaction Type.
- In the PowerBI Dashboard; Move to individual customer behavior, transaction frequency, total transaction value per customer, high-activity/high-value customers and normalized segment performance. This will show whether total activity is driven by segment size or by stronger customer-level behavior.
- Enhance the existing dashboard rather than rebuild it. Add decision-oriented KPIs, drill-through or drill-down, stronger comparisons, clearer navigation and a more coherent management story that helps users investigate why a KPI or pattern looks the way it does.

Remaining limitations.

- The data is synthetic and covers only 3 months.
- The risk flag is not a fraud label.
- Engagement differences are small.
- Findings show association, not causation.

What must be completed in Week 4.

- Final testing, refinement, final documentation and presentation prep.